

Whole Mount Subcutaneous White Adipose Tissue Isolation and Dissection

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1. Sacrifice mice according to IACUC approved protocol. Collect body weight.
2. Place mouse ventral side up, spray with 70% ethanol.
3. Make incision extending from top of the sternum to base of the tail making sure not to cut through peritoneum (fig 1a - black line).
 - a. Note: additional cuts can be made (fig 1a - red lines) for obese mice to assist in separation of skin from the body with less distress to the more fragile obese tissue.
4. Gently use thumbs to peel ventral skin away from both flanks being careful not to damage tissue (Fig 1b).
5. Identify the inguinal scWAT depot against the skin beside each hip and use scissors to cut large vasculature leading to the depot (fig 1b).
6. Gently pull depot away from skin cutting any fibrous connective tissue holding it against the skin being sure not to cut the depot.
7. Once depot has been extracted weigh it and then place it in 2% PFA for 4-16hrs for fixation and subsequent immunostaining.
 - a. Note: A fat depot ~0.20g needs only 4hrs, 0.50g needs approx. 8hrs, ~1.0g 16hrs.
8. Repeat for inguinal scWAT on the other side.

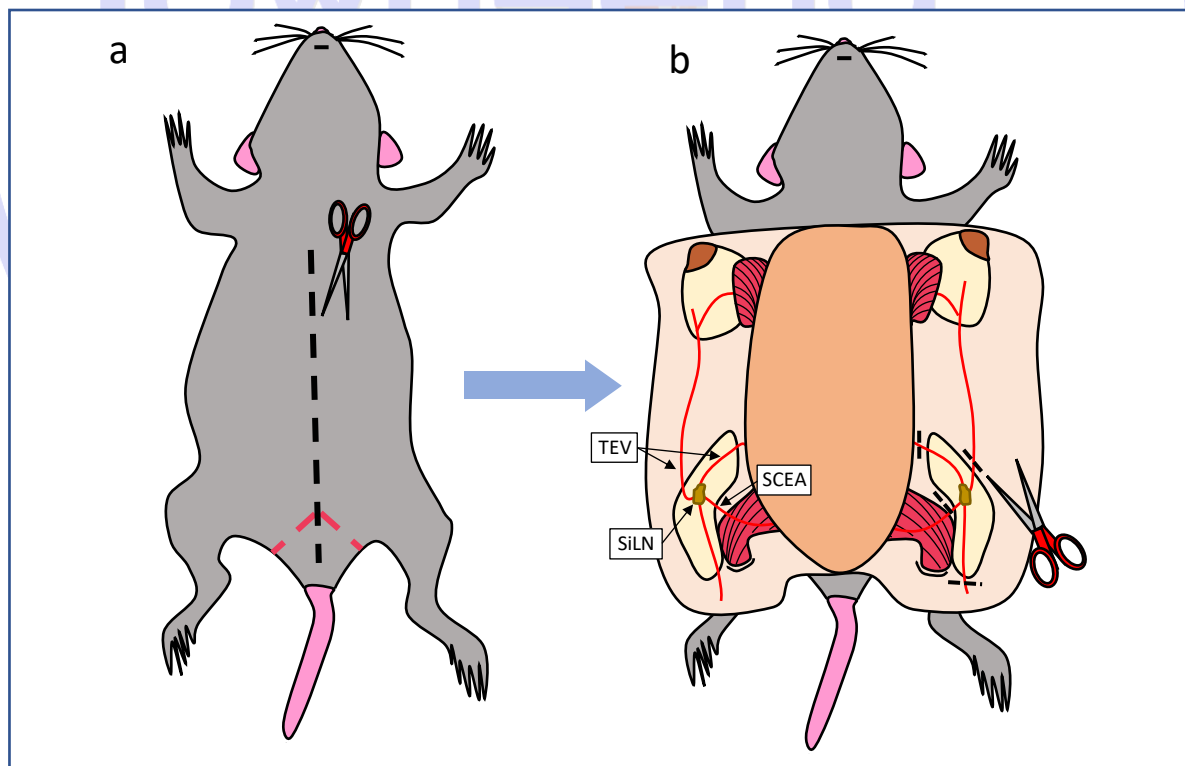


Figure 1. Ventral Incision and inguinal scWAT depot extraction. Anatomical features pointed out for reference: subiliac lymph node (SiLN), thoracoepigastric vein (TEV), and superficial caudal epigastric artery (SCEA).

9. Identify the axillary scWAT depot under both arms
10. Use forceps to pull away a thin piece of muscle that partially covers the depot (fig 2a)
11. Use scissors to cut at the edges of the tissue where it is held down by fibrous connective tissue (fig 2b).
12. There is a small piece of BAT at the top of the axillary depot that can be left attached for orientation.
13. Gently pull away depot from skin only cutting the fibrous connective tissue underneath.
14. Once depot has been extracted weigh it and then place it in 2% PFA for 4-16hrs for fixation and subsequent immunostaining.
 - a. Note: A fat depot ~0.20g needs only 4hrs, 0.50g needs approx. 8hrs, ~1.0g 16hrs.
15. Repeat for axillary scWAT on the other side.

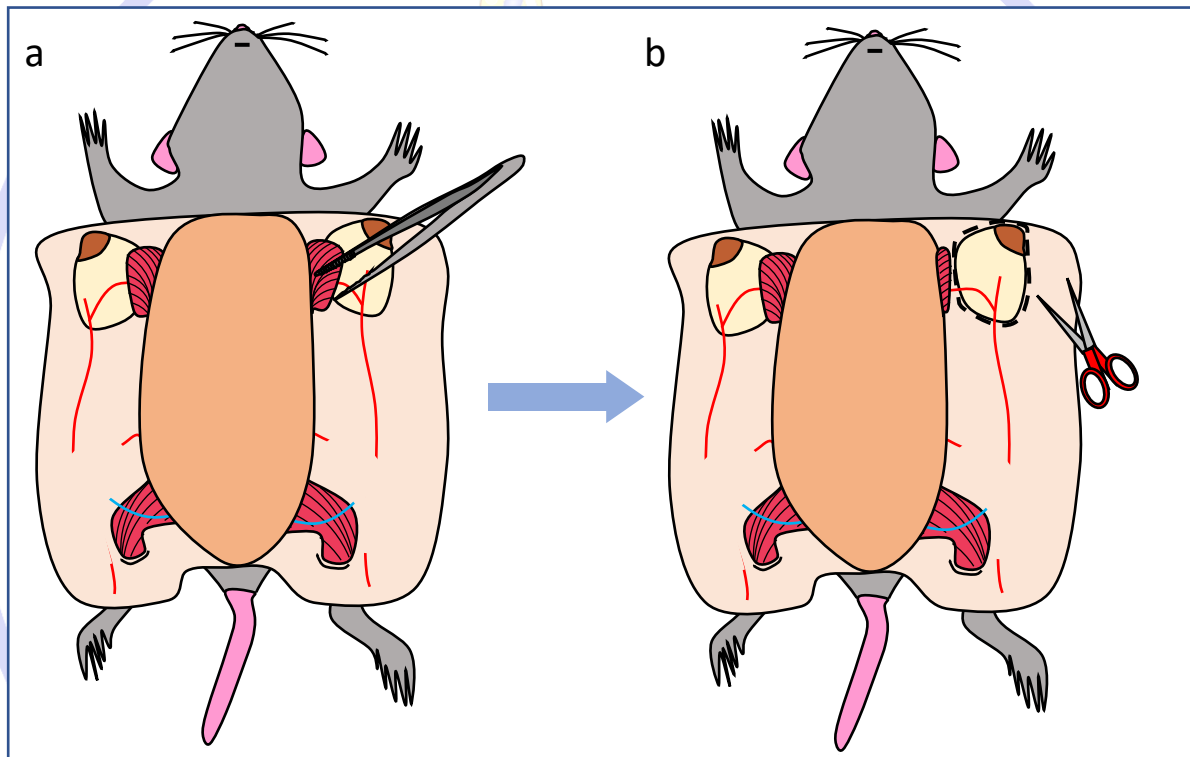


Figure 2. Extraction of axillary scWAT depot.